Baseline as a Covariate in the Analysis of the Hybrid N-of-1 Design: Advantages and Disadvantages

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Purpose. The published analysis of Hendrickson et al. [1], and every analysis in this project so far, treats the baseline symptom score as an observation of the response, a ninth row coded off drug. This paper weighs the alternative used in most confirmatory trials: baseline as a covariate in a model for the post-baseline visits. It also covers a third option, constrained longitudinal data analysis, which keeps baseline in the response but changes how it is modeled. Section 4 gives a formal account of what adjusting for baseline does to each quantity the analysis estimates, with each result checked numerically against the published construct. No new Monte Carlo simulation was run; Section 9 lists what the paused covariance study (`08-covariance-study.R`) would settle.

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Notation index and glossary

Notation follows `analysis/report/NOTATION.md` and `docs/40`. Symbols marked *local* are defined here.

Table 1: Notation index and glossary of symbols used in this document

Symbol	Meaning	Status
Y_{it}	symptom score of participant i at visit t (code `Sx`); $t = 0$ is baseline	canonical
BL_i	baseline symptom score, $Y_{i0} = \mathrm{BL}_i$	canonical
BR_{it}, PB_{it}, TV_{it}	drug, expectancy and natural-course responses	canonical
S_{it}	sum of the three responses, $TV_{it} + PB_{it} + BR_{it}$	local
B_i	biomarker	canonical
D_{it}	binary drug state (code `Db`); 0 at baseline	canonical
β_B	biomarker main effect	local
β_D, $\beta_{bm:D}$	drug main effect; biomarker-by-drug interaction	canonical
ε_{it}, u_i	residual; participant random intercept, with variances σ_e^2, σ_u^2	canonical; variances local
c_{bm}, σ_{BR}, σ_{bm}	biomarker-BR coupling; SDs of BR and of the biomarker	canonical
g_t	coupling of the biomarker to BR at visit t (`docs/40`, Section 5)	local
$t_{1/2}$	carryover half-life	canonical
$f(t)$	time adjustment in the mean model	local
$\mathbf{Y}_{ip}$, X_{ip}, β	vector of the 8 post-baseline outcomes; their design matrix; fixed effects	local
μ_0	baseline mean intercept	local
σ_{00}, σ_{p0}, Σ_{pp}	baseline variance, baseline-by-visit covariances, post-baseline covariance	local
σ_{tt}, σ_{t0}	variance at visit t; its covariance with baseline	local
$\Sigma_{pp\cdot 0}$	post-baseline covariance conditional on baseline	local
J	matrix of ones	local
ϕ, ϕ_t, $\bar{\phi}$	baseline regression slopes, $\phi = \sigma_{p0}/\sigma_{00}$; slope at visit t; common slope	local (renamed from γ, which NOTATION reserves)

Symbol	Meaning	Status
ϕ_{BL}	common baseline slope in ANCOVA	local
w	generic baseline weight in Proposition 3	local
$\mathbf{a}$	contrast weights on the post-baseline visits, $\sum_t a_t = 0$	local
Δ_i, Δ_i^{E9}, Δ_i^{E9b}	participant contrast $\mathbf{a}'\mathbf{Y}_{ip}$; its E9 and E9b forms	local
G	variance reduction from conditioning a contrast on baseline	local
$r^2_{\Delta,\mathrm{BL}}$	squared correlation of Δ_i with baseline	local (renamed from ρ^2, which NOTATION reserves for serial correlation)
s_i, d_{it}, σ_s^2	shared response level; visit-specific deviation; variance of s_i	local
$\bar{S}_{i,\mathrm{on}}$, $\bar{S}_{i,\mathrm{off}}$, $\bar{g}_{\mathrm{on}}$, $\bar{g}_{\mathrm{off}}$	on- and off-drug means of S_{it} and of g_t	local
n_{on}, n_{off}	numbers of on-drug and off-drug post-baseline visits in a path	local
V, A, M	working covariance; bread and meat of the sandwich variance	local

- **Response coding.** Baseline enters as a row of the outcome, at $t = 0$ with $D = 0$ (the published analysis).
- **Covariate coding (ANCOVA).** Only post-baseline visits are outcomes; BL_i is a fixed covariate.
- **cLDA.** Constrained longitudinal data analysis [5, 6]: baseline stays in the response vector, its mean is constrained to be common across randomization groups, and the covariance of all visits is modeled, usually unstructured.
- **E9 and E9b.** The paired-difference statistics of `docs/37`: each participant's on-drug mean minus off-drug mean, regressed on the biomarker. E9 uses post-baseline visits; E9b also counts baseline as an off-drug visit, which mirrors the response coding.
- **Unstructured covariance (`us`).** Every visit has its own variance and every pair its own covariance; the MMRM default.

1. Summary

1. **Baseline adjustment changes the analysis, not the data.** The same replicates can be fitted both ways (Section 2).
2. **In this construct the baseline slopes are all exactly 1.** Post-baseline scores are $\mathrm{BL}_i - S_{it}$, and baseline is uncorrelated with every response component and with the biomarker. Conditioning on baseline is therefore change from baseline (Section 3, Proposition 1).
3. **A covariate gives a within-participant contrast nothing when the baseline slopes are equal.** The variance reduction from conditioning a contrast $\mathbf{a}'\mathbf{Y}_{ip}$ on baseline is $\sigma_{00}(\mathbf{a}'\phi)^2$. That is zero whenever every visit has the same slope, because the weights sum to zero. The interaction is such a contrast. The construct's E9 variance is 44.13 with or without conditioning (Proposition 2).
4. **Between participants the gain is large.** Conditioning reduces the variance of a blinded post-baseline value from 599 to 257. That benefits the biomarker main effect and comparisons of path means, which are not the target (Proposition 3).
5. **What matters for the interaction is whether the baseline row is counted as an off-drug visit.** Counting it leaves part of each participant's shared response level in the contrast. That raises the contrast variance by 15% to 16%. It also creates a residual interaction of $1/(n_{\mathrm{off}}+1)$ of the full value under the published step rule with carryover: the power floor of the published analysis (Proposition 4).
6. **The baseline row is incompatible with a random-intercept covariance.** It has no noise of its own, which compound symmetry cannot represent. Dropping it, or modeling an unstructured covariance, removes that misspecification of the model-based standard errors (Proposition 5).
7. **Recommendation.** Analyze the post-baseline visits with baseline as a covariate in an MMRM with an unstructured covariance, or use cLDA, which coincides with it for complete data. Avoid the published combination of a baseline row, a random intercept and linear time. The covariate's value for the interaction is robustness, not efficiency: it pays when baseline predicts on-drug and off-drug visits differently, which the construct rules out but a real trial may not (Sections 4 and 8). *Superseded (2026-10-07):* see the dated note in Section 8; the project's current primary analysis is F1 of `docs/46` and `docs/49`.

2. The two codings

Response coding (published). For each participant the outcome vector has nine entries, $(\mathrm{BL}_i, Y_{i1}, \ldots, Y_{i8})$. The baseline row has $t = 0$, $D = 0$ and expectancy 0 (`lme_analysis.R` at commit `58b32a9`, lines 62 to 87). The published model, `lmer(Sx ~ bm + Db + t + bm * Db + (1 | ptID), data = long)`, treats it as one more off-drug visit.

Covariate coding. The outcome vector has the eight post-baseline entries, and baseline is a covariate:

$$Y_{it} = \beta_0 + \phi_{\mathrm{BL}}\,\mathrm{BL}_i + \beta_B B_i + \beta_D D_{it} + f(t) + \beta_{bm:D}\,B_i D_{it} + \varepsilon_{it}, \qquad t = 1, \ldots, 8,$$

with $f(t)$ the time adjustment and ε_i given a working covariance, for example `us(visit | ptID)` in `mmrm`.

The simulated data are unchanged: the same replicates, with the baseline value moved from a row of the response to a column of the design matrix.

3. The data-generating process

In the published construct $Y_{it} = \mathrm{BL}_i - S_{it}$ for $t \geq 1$, and BL_i is drawn independently of the biomarker and of all response components. Hence

$$\sigma_{t0} = \mathrm{Cov}(Y_{it}, \mathrm{BL}_i) = \mathrm{Var}(\mathrm{BL}_i) = \sigma_{00} \quad \text{for every } t \geq 1, \qquad \Sigma_{pp} = \sigma_{00} J + \mathrm{Cov}(\mathbf{S}_{ip}),$$

with J the matrix of ones. Computed exactly from the construct (path 1, no interaction; `docs/40`, Section 6):

Table 2: Exact variances and covariances by phase, path 1, no interaction

	Variance	Covariance within phase	Covariance with baseline
Baseline	342		
Open label (weeks 4, 8)	710	605	342
Blinded (weeks 9 to 20)	599	527	342

with 556 between open-label and blinded visits. Two features matter below. Baseline has no noise of its own: its covariance with every later visit equals its own variance. And every later visit has the same covariance with baseline.

4. A formal assessment of adjusting for baseline

Throughout, the outcomes are multivariate normal, the mean model is correctly specified, and the data are complete. Write $\mathbf{Y}_i = (Y_{i0}, \mathbf{Y}'_{ip})'$ with

$$E(Y_{i0}) = \mu_0 + \beta_B B_i, \qquad E(\mathbf{Y}_{ip}) = X_{ip}\beta, \qquad \Sigma = \begin{pmatrix} \sigma_{00} & \sigma'_{p0} \\ \sigma_{p0} & \Sigma_{pp} \end{pmatrix}.$$

Baseline is off drug, so $D_{i0} = 0$ and neither β_D nor $\beta_{bm:D}$ enters its mean.

Proposition 1: the covariate model is the conditional model

Conditional on baseline,

$$\mathbf{Y}_{ip} \mid Y_{i0} \;\sim\; N\Big(X_{ip}\beta + \phi\,(Y_{i0} - \mu_0 - \beta_B B_i),\;\; \Sigma_{pp\cdot 0}\Big), \qquad \phi = \frac{\sigma_{p0}}{\sigma_{00}}, \quad \Sigma_{pp\cdot 0} = \Sigma_{pp} - \frac{\sigma_{p0}\sigma_{p0}'}{\sigma_{00}}.$$

Proof. The standard conditional distribution of a partitioned multivariate normal. □

ANCOVA with visit-specific baseline slopes ϕ_t and residual covariance $\Sigma_{pp\cdot 0}$ is exactly this model. A common slope ϕ_{BL} is the special case $\phi_t \equiv \phi_{\mathrm{BL}}$. Conditioning changes the intercept and the biomarker main effect, which absorbs $-\phi_t\beta_B$. It leaves β_D and $\beta_{bm:D}$ unchanged, because they do not appear in the baseline mean. The estimand of the interaction is the same under both codings.

In the construct, $\sigma_{t0} = \sigma_{00}$ for every t, so $\phi = \mathbf{1}$ (verified: every slope is 1 to six decimals) and $\Sigma_{pp\cdot 0} = \mathrm{Cov}(\mathbf{S}_{ip})$: conditioning on baseline removes exactly its variance from every entry. The covariate analysis then coincides with a change-from-baseline analysis, and the classic argument between the two [2, 3] does not arise.

Proposition 2: a covariate adds nothing to a within-participant contrast when the slopes are equal

Let $\mathbf{a}$ be contrast weights on the post-baseline visits with $\sum_t a_t = 0$; the E9 weights $1/n_{\mathrm{on}}$ and $-1/n_{\mathrm{off}}$ are an example. Then

$$\mathrm{Var}(\mathbf{a}'\mathbf{Y}_{ip} \mid Y_{i0}) = \mathbf{a}'\Sigma_{pp}\mathbf{a} - \frac{(\mathbf{a}'\sigma_{p0})^2}{\sigma_{00}} = \mathbf{a}'\Sigma_{pp}\mathbf{a} - \sigma_{00}\,(\mathbf{a}'\phi)^2.$$

The reduction $G = \sigma_{00}(\mathbf{a}'\phi)^2$ is zero whenever the slopes are equal, $\phi_t \equiv \bar{\phi}$, because then $\mathbf{a}'\phi = \bar{\phi}\sum_t a_t = 0$.

Proof. Proposition 1 and $\mathrm{Var}(\mathbf{a}'\mathbf{Z}) = \mathbf{a}'\,\mathrm{Cov}(\mathbf{Z})\mathbf{a}$. □

Consequence for the interaction. The E9 slope regresses $\Delta_i = \mathbf{a}'\mathbf{Y}_{ip}$ on B_i. Adding baseline to that regression reduces the residual variance by the factor $1 - r^2_{\Delta,\mathrm{BL}}$, where $r^2_{\Delta,\mathrm{BL}} = (\mathbf{a}'\sigma_{p0})^2/(\sigma_{00}\,\mathbf{a}'\Sigma_{pp}\mathbf{a})$. With equal slopes this is zero, and the extra parameter costs one degree of freedom for nothing.

In the construct $\mathbf{a}'\sigma_{p0} = 0$ exactly, and the variance of the E9 contrast is the same with and without conditioning: 44.13 for paths 1 and 2, 45.03 for paths 3 and 4 (verified).

When the covariate does help a within-participant contrast. $\mathbf{a}'\phi \neq 0$ requires baseline to predict on-drug visits differently from off-drug visits. Two mechanisms give that:

- **Baseline severity modifies the drug response.** Then ϕ_t is larger at on-drug visits, and the appropriate model also includes $\mathrm{BL}_i \times D_{it}$.
- **Baseline predicts a time-varying natural course.** An example is regression to the mean that is strongest early in the trial. Then ϕ_t changes with time, and in the Hybrid design drug state is confounded with time: on drug early, off drug late. So $\mathbf{a}'\phi \neq 0$.

Measurement error in baseline alone does not qualify. It attenuates every slope by the same reliability factor, so the slopes stay equal and G stays zero.

Proposition 3: between participants, the classic ANCOVA gain

For a comparison of post-baseline means between participants, such as the biomarker main effect or a difference between paths, the relevant variance is that of a post-baseline value. For a single visit t, the three classic analyses [2, 3] have

$$\mathrm{Var}(Y_{it}) = \sigma_{tt}, \qquad \mathrm{Var}(Y_{it} - Y_{i0}) = \sigma_{tt} + \sigma_{00} - 2\sigma_{t0}, \qquad \mathrm{Var}(Y_{it} \mid Y_{i0}) = \sigma_{tt} - \frac{\sigma_{t0}^2}{\sigma_{00}},$$

and the conditional variance is never larger than the other two. It equals the change-score variance exactly when $\phi_t = 1$.

Proof. $\mathrm{Var}(Y_{it} - w\,Y_{i0})$ is a quadratic in w, minimized at $w = \phi_t$, with minimum $\sigma_{tt} - \sigma_{t0}^2/\sigma_{00}$. $w = 0$ gives the first expression and $w = 1$ the second. □

In the construct, for a blinded visit: 599 unadjusted, 257 as change or conditional on baseline (verified). That is a 57% reduction, the largest virtue of baseline adjustment. It accrues to between-participant terms, which in this design are the biomarker main effect and the path means. Neither is the target.

Proposition 4: the baseline row as an off-drug visit

The response coding counts baseline as an off-drug observation. For a path's paired-difference contrast, that is the E9b statistic, with off-drug mean $(Y_{i0} + \sum_{\mathrm{off}} Y_{it})/(n_{\mathrm{off}} + 1)$. Substituting $Y_{i0} = \mathrm{BL}_i$ and $Y_{it} = \mathrm{BL}_i - S_{it}$:

$$\Delta_i^{\mathrm{E9}} = -\bar{S}_{i,\mathrm{on}} + \bar{S}_{i,\mathrm{off}}, \qquad \Delta_i^{\mathrm{E9b}} = -\bar{S}_{i,\mathrm{on}} + \frac{n_{\mathrm{off}}}{n_{\mathrm{off}} + 1}\,\bar{S}_{i,\mathrm{off}}.$$

Baseline cancels from both. The only difference is that E9b shrinks the off-drug mean of the responses toward zero, the value they have at baseline. Two consequences follow.

(a) The interaction slope. With g_t the coupling of the biomarker to the drug response at visit t (`docs/40`, Section 5),

$$\beta^{\mathrm{E9}}_{bm:D} = -c_{bm}\frac{\sigma_{BR}}{\sigma_{bm}}(\bar{g}_{\mathrm{on}} - \bar{g}_{\mathrm{off}}), \qquad \beta^{\mathrm{E9b}}_{bm:D} = -c_{bm}\frac{\sigma_{BR}}{\sigma_{bm}}\Big(\bar{g}_{\mathrm{on}} - \frac{n_{\mathrm{off}}}{n_{\mathrm{off}}+1}\,\bar{g}_{\mathrm{off}}\Big).$$

- **Without carryover** ($\bar{g}_{\mathrm{off}} = 0$) the two coincide.
- **Under the published step rule with any carryover** ($\bar{g}_{\mathrm{on}} = \bar{g}_{\mathrm{off}} = 1$), the E9 slope is zero. The E9b slope is $1/(n_{\mathrm{off}}+1)$ of the no-carryover value, produced entirely by the uncoupled baseline row. That is the residual power floor of the published analysis (0.081 in the closed form; 0.080 to 0.108 for `lmer`, `docs/37`, Section 6; the range was earlier given here as 0.08 to 0.10). It measures a pre-treatment comparison, not an interaction during the trial.

(b) The precision. Write $S_{it} = s_i + d_{it}$, where s_i is the participant's shared response level, with variance σ_s^2 equal to the common covariance of the responses across visits. E9 removes s_i exactly, since its weights on the responses sum to zero. E9b leaves $-s_i/(n_{\mathrm{off}}+1)$ in the contrast, which adds $\sigma_s^2/(n_{\mathrm{off}}+1)^2$ to its variance. The changes in the visit-specific terms, whose off-drug mean is shrunk by $n_{\mathrm{off}}/(n_{\mathrm{off}}+1)$, partly offset this. In the construct $\sigma_s^2 \approx 200$, so the shared level adds about 12.5 ($n_{\mathrm{off}} = 3$) and 8.0 ($n_{\mathrm{off}} = 4$), and the net change is smaller (derived; the earlier text called the offset "smaller changes", but it removes about 45% and 16% of the addition). The contrast variance rises from 44.13 to 51.04 (paths 1 and 2, $n_{\mathrm{off}} = 3$) and from 45.03 to 51.77 (paths 3 and 4, $n_{\mathrm{off}} = 4$): increases of 16% and 15% (verified, rerun of `02-baseline-propositions.R` on 2026-10-07). In the closed form this costs 0.06 to 0.07 in power (Section 6).

Proposition 5: the working covariance and the validity of the standard errors

For a working covariance V, the GLS estimator $\hat{\beta}_V = \left(\sum_i X_i' V^{-1} X_i\right)^{-1} \sum_i X_i' V^{-1} \mathbf{Y}_i$ is unbiased under a correct mean model, with

$$\mathrm{Var}(\hat{\beta}_V) = A^{-1} M A^{-1}, \qquad A = \sum_i X_i' V^{-1} X_i, \quad M = \sum_i X_i' V^{-1} \Sigma V^{-1} X_i.$$

The model-based variance reported by the software is A^{-1}. It is correct when $V \propto \Sigma$, and otherwise can be too large or too small. The sandwich estimator of $A^{-1} M A^{-1}$ is consistent for any V.

Proof. Standard; $\hat{\beta}_V$ is linear in $\mathbf{Y}_i$ with $E(\mathbf{Y}_i) = X_i\beta$. □

In the construct, the published random intercept implies $V = \sigma_u^2 J + \sigma_e^2 I$ over the nine rows. Every row of V then carries the same residual variance $\sigma_e^2 > 0$, whereas the baseline row of Σ has none ($\sigma_{00} = \sigma_{t0}$). No choice of (σ_u^2, σ_e^2) matches it, so the model-based standard errors are not guaranteed correct. The residual mismatch between open-label and blinded visits remains. Dropping the baseline row (covariate coding) or modeling all nine rows with an unstructured covariance (cLDA) removes the baseline part of the mismatch. Only the unstructured covariance, or a sandwich estimator, removes all of it.

What this means for each quantity

Table 3: Effect of baseline as a covariate on each estimated quantity

Quantity	Effect of baseline as a covariate	Result
Interaction $\beta_{bm:D}$	no efficiency gain from the covariate when slopes are equal; gains from no longer counting baseline as off drug	Propositions 2 and 4
Drug main effect β_D	no gain for its within-participant part; loses baseline's off-drug information	Propositions 2 and 4
Biomarker main effect β_B	large gain (57% variance reduction per visit)	Proposition 3
Model-based standard errors	removes the baseline row's incompatibility with compound symmetry	Proposition 5
Robustness	protects the interaction when baseline predicts on and off visits differently	Proposition 2

5. Advantages of baseline as a covariate

1. **It removes the row the published covariance cannot represent** (Proposition 5). The baseline row has the smallest variance (342), no noise of its own, and lower correlation with later visits (0.69 to 0.76, against 0.85 to 0.88 among later visits). That is the largest departure from compound symmetry in the outcome covariance.
2. **It stops a pre-treatment value from diluting the on-off contrast** (Proposition 4). Without carryover this improves the contrast's precision by 15% to 16% in variance. Under the step rule it removes a residual interaction that does not describe the trial.
3. **It protects the interaction when baseline is informative about the course of response** (Proposition 2). A covariate removes the part of the contrast predictable from baseline when $\mathbf{a}'\phi \neq 0$: for example, when severity moderates the drug response or predicts a time-varying natural course. The construct excludes both; real trials may not.

4. **It estimates the baseline coefficient instead of fixing it.** With measurement error or regression to the mean, the slopes fall below
    1. A covariate estimates them; the response coding under a random intercept implicitly ties baseline to the same participant level as every later visit. ANCOVA is the standard choice for this reason [2, 3].
5. **It is the convention reviewers expect.** Baseline-adjusted analysis of post-baseline visits, usually as an MMRM, is the default in confirmatory trials, and regulatory guidance on covariate adjustment assumes it [4].
6. **It opens a direct test of severity as a moderator.** A $\mathrm{BL}_i \times D_{it}$ term is the natural extension. In the construct it is exactly zero.

6. Disadvantages of baseline as a covariate

1. **It gives the interaction no efficiency gain in this construct** (Proposition 2). Any power change comes from dropping the baseline row, not from the covariate, and costs one degree of freedom.
2. **It discards baseline's off-drug information for the drug main effect.** Baseline is the only off-drug value with no expectancy, carryover or natural-course change. How much that matters depends on the time adjustment: with visit effects, baseline has its own mean and carries no drug information anyway (`docs/42`).
3. **It changes the published estimand.** Results are no longer directly comparable with Hendrickson et al.'s, including the power floor under the step rule (Proposition 4a).
4. **A participant without a baseline value is lost** unless it is imputed. cLDA is more efficient than ANCOVA when baselines are missing [5, 6]. In this design baseline precedes randomization and should rarely be missing.
5. **The covariate does not fix the post-baseline covariance** (Proposition 5). Open-label and blinded visits still differ (variance 368 against 257 after conditioning, covariance 263 against 185). A random intercept on the post-baseline visits remains misspecified; the covariate coding is fully effective only with `csh` or `us`.

Closed-form evidence on the baseline row (E9 against E9b; Hybrid, compound symmetry, $N = 70$, strength 0.25; verified, `docs/37`, Sections 4 and 6):

Table 4: Power and SD of E9 and E9b by carryover rule and $t_{1/2}$

Rule, $t_{1/2}$	E9 power	E9b power	E9 SD	E9b SD
Step, 0	0.721	0.654	0.0507	0.055
Step, 0.1 or more	0.050	0.081	0.0531	0.057
Mean moderation, 0 to 1	0.681	0.619	0.0531	0.057

These are results for the paired-difference statistic. They show the direction and size of Proposition 4's effects. A mixed model weights the visits by its working covariance, so its numbers depend on that covariance (Section 7).

7. Constrained longitudinal data analysis: when the choice matters

Liang and Zeger proposed keeping baseline in the response vector, constraining its mean to be common across randomization groups, and modeling the covariance of all visits [5].

With complete data and an unstructured covariance, cLDA and ANCOVA with visit-specific slopes give the same estimates of the post-baseline parameters. The joint likelihood factorizes as $f(Y_{i0}) \times f(\mathbf{Y}_{ip} \mid Y_{i0})$. Under an unstructured Σ, the parameters of the conditional factor (β, ϕ, $\Sigma_{pp\cdot 0}$) vary independently of those of the marginal factor (μ_0, σ_{00}), apart from any parameter shared by the two means (Proposition 1). In the Hybrid design every path is off drug at baseline with the same time and expectancy, so randomization's constraint holds automatically. The only shared mean parameter is the biomarker main effect β_B, and allowing it to differ at baseline restores exact equivalence. The equivalence is a property of the likelihood; it has not been simulated here. Lu [6] shows that cLDA is more efficient when baselines are missing.

The choice matters most under the published model. With a random intercept and linear time, the baseline row is not modeled by its own mean or covariance. It enters as an off-drug visit at $t = 0$ under a covariance it does not have (Proposition 5), it dilutes the on-off contrast (Proposition 4), and it anchors the time trend.

Table 5: Effect of the baseline coding under each analysis model

Analysis	Effect of the baseline coding
Random intercept, linear `t` (published)	large: the row is misfit, dilutes the contrast and has leverage
Random intercept, visit factor	moderate: the row has its own mean, but the covariance is still wrong
Unstructured covariance, visit factor	none for complete data: cLDA and ANCOVA coincide

Note (2026-10-07). The last row holds only when the row model frees the biomarker main effect β_B at baseline, as the paragraph above states. With the published row coding, which shares β_B between baseline and the post-baseline off-drug visits, the row and covariate analyses differ, and under AR(1) data the shared coefficient is the source of the row coding's extra power (`docs/45`, Sections 6 and 10).

8. Recommendation

For a Hybrid trial analysis. Analyze the eight post-baseline visits with baseline as a covariate, in an MMRM with an unstructured covariance and visit effects. Allow visit-specific baseline slopes if their variation is plausible, and consider $\mathrm{BL}_i \times D_{it}$ as a secondary moderator. With complete baselines this coincides with cLDA, which is the more efficient form when baselines may be missing. The covariate's value for the interaction is robustness against baseline predicting the course of response (Proposition 2), not efficiency. Avoid the published combination of a baseline row, a random intercept and linear time.

For simulations that reproduce Hendrickson et al. Keep the response coding, since it defines the published estimand. Report the covariate coding alongside it wherever the comparison bears on conclusions.

Superseded (2026-10-07). The trial-analysis recommendation above was first revised by `docs/45` (Section 9), which made the row coding with an unstructured covariance and Kenward-Roger inference primary and the covariate coding a sensitivity analysis. That primary was in turn rejected by the pre-registered stress test (`docs/46`, script `11-strawman-stress-test.R`; 41 cells, 28,000 trials, 14 analyses), where it is analysis A: pooled null rejection 0.057 and 0.087 at $N = 35$, and 0.195 to 0.416 in the three pre-registered prognostic cells. The analysis closest to this section's recommendation, EA (the same model with β_B freed at baseline, which `docs/45`, Section 9.2, treats as equivalent to the covariate coding), held its size (pooled 0.052, no cell above 0.070) and was nominal in every prognostic cell, but had the largest maximin shortfall of the survivors (0.318). The decision rule, rerun with all 14 analyses, selects F1, a random intercept with `corCAR1` residuals, CR2 inference, the baseline row, and the expectancy term with its biomarker interaction (`docs/49`; verified against `decision-screen-ds-nosym-all14.csv` and `decision-maximin-ds-nosym-all14.csv`). The stress test used linear time, not the visit effects proposed here.

9. Open questions

The paused covariance study would answer these if the covariate coding were added: the post-baseline `cs` and `us` analyses with each time adjustment, about 40% more run time.

1. **Size.** Does the covariate coding remove the anticonservatism of the flexible time adjustments in Hybrid (`docs/40`, Section 9)? Proposition 5 predicts that it removes part of the cause.
2. **Power.** Does the mixed-model gain from dropping the baseline row match Proposition 4's 15% to 16% variance reduction?
3. **Equivalence.** Do cLDA and ANCOVA coincide under an unstructured covariance, as Section 7 derives?

4. **Robustness.** Under a construct in which baseline predicts the natural course or moderates the drug response, how large is Proposition 2's gain? That requires extending the construct; the software has no baseline-to-component link (`docs/31`, Section 18.1).

Status (2026-10-07). The covariance study was completed and reported in `docs/45`. Question 1: no; in the one Hybrid cell with a flexible-time excess, the covariate coding with Kenward-Roger rejects 0.060 (spline) and 0.064 (visit factor), against 0.068 and 0.072 for the row coding, and only CR2 removes the excess (`docs/45`, Section 4.4). Question 2 was not isolated there, because the row coding's power also depends on the shared β_B (`docs/45`, Section 6). Question 3: the two coincide only when β_B is freed at baseline (`docs/45`, Section 6). Question 4 remains open: the stress test of `docs/46` added links of the biomarker to TV, to PB and to baseline severity (its cells 40 and 41), but no link from baseline to the response components (inspected, `11-strawman-stress-test.R`).

10. Limitations and evidence status

- **Derivations and numeric checks, no Monte Carlo.** Propositions 1 to 5 are standard multivariate-normal and GLS results applied to this design (derived). Their numeric consequences were computed exactly from the construct's implied covariance (verified, `analysis/scripts/quick-sim/hybrid-design-primer/02-baseline-propositions.R`). The power figures are closed-form results for E9 and E9b, validated by Monte Carlo in `docs/37` (verified). Statements about regression to the mean and measurement error come from the literature [2, 3].
- **One construct.** The published compound-symmetry construct has no baseline measurement error, no regression to the mean and no link between baseline and the response. Each would strengthen the case for the covariate (Proposition 2).
- **One design.** Hybrid at $N = 70$. In CO the baseline row is equally uncoupled, but the paths differ in drug state from the first visit.

11. References

1. Hendrickson RC, Thomas RG, Schork NJ, Raskind MA. Optimizing aggregated N-of-1 trial designs for predictive biomarker validation: statistical methods and theoretical findings. *Frontiers in Digital Health* 2020; 2:13. doi:10.3389/fdgth.2020.00013.
2. Vickers AJ, Altman DG. Statistics notes: analysing controlled trials with baseline and follow up measurements. *BMJ* 2001; 323(7321):1123-1124. doi:10.1136/bmj.323.7321.1123.
3. Senn S. Change from baseline and analysis of covariance revisited. *Statistics in Medicine* 2006; 25(24):4334-4344. doi:10.1002/sim.2682.

4. European Medicines Agency. Guideline on adjustment for baseline covariates in clinical trials. EMA/CHMP/295050/2013, 2015. U.S. Food and Drug Administration. Adjusting for covariates in randomized clinical trials for drugs and biological products: guidance for industry, 2023.
5. Liu GF, Lu K, Mogg R, Mallick M, Mehrotra DV. Should baseline be a covariate or dependent variable in analyses of change from baseline in clinical trials? *Statistics in Medicine* 2009; 28(20):2509-2530. doi:10.1002/sim.3639. (Describes the cLDA model of Liang KY and Zeger SL, *Sankhya Series B* 2000; 62:134-148.)
6. Lu K. On efficiency of constrained longitudinal data analysis versus longitudinal analysis of covariance. *Biometrics* 2010; 66(3):891-896. doi:10.1111/j.1541-0420.2009.01332.x.